NU MOA
School of Information Technology
BS Information Technology

PRE-PROPOSAL CAPSTONE TOPIC FORM

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TITLE OF PROPOSED CAPSTONE PROJECT *The title should briefly and simply reflect the main topic of the project.*	
Travex: A Package Booking, Smart Recommendation System, and Customer Management System for M&RC Travel and Tours	
SUSTAINABLE DEVELOPMENT GOAL (SDG)	**SDG 8: Decent Work and Economic Growth** **SDG 9: Industry, Innovation and Infrastructure**
CHOOSE SPECIALIZATION:	☒ Mobile and Web Applications (MWA)
	☐ Multimedia Arts and Animation (MAA)
	☐ General Course
CAPSTONE PROJECT DISCIPLINE:	Mobile and Web Software Development

1. PROJECT CONTEXT

Please state the rationale of your proposed Capstone Project

The Travel and Tourism industry have been transformed by information and communication technologies worldwide, and it is one of the biggest economic sectors across the globe. Information and communication technologies increased the competition for tourism-related organizations and businesses to adopt the latest technologies in order to achieve competitive edge and growth. The travel and leisure sector applied information technologies to cut costs, save on labour, increase operational efficiency, and improve customer service and experience.

Travel Agency companies or businesses are nowadays emerging, offering travel and tour packages that includes transportation, accommodation, and other services involving visiting tourist spots and recreational activities that are bundled together. Travel agencies on the market attract customers by selling services such as accommodation, airfare, tour activities, and other touristic services which can be purchased completely in one channel.

In recent years, customers have been relying on travel agents for itinerary planning and other arrangements, and they undergo a traditional process of visiting physical locations and performing business operations,

which can be considered time-consuming and inefficient for both customers and the agency. Thus, developing a mobile and web-based travel and tour booking, smart recommendation and customer management system aims to benefit target enterprises through having advantages such as convenience for the customers, reducing storefront costs, greater reach and access for consumers, better customer service, efficiency in handling bookings and managing payment transactions.

Implementing this information system allows travel and tour agencies to be flexible and give customers more information about packages and services that they offer, such as the detailed inclusions, exclusions and itineraries of the tour package. Moreover, online booking platforms have become an increasingly popular means for tourism, it helps consumers to make more reasonable choices and allows them to make booking of any preferred time, reduce workload, and handling of manual transactions. The Package Booking, Smart Recommendation and Customer Management System provides key features for customers including browsing travel destinations, personalized recommendations, flexible booking options, review and rating packages, and quality customer service. Moreover, it aims to be cost-effective and time-efficient for the agency since it will digitalize business operations and processes such as staff and admin managing and monitoring available tour packages, user accounts, bookings, and transactions. It also integrates a chatbot that will provide customer support when it comes to addressing inquiries and answering FAQs which leads to reducing staff burden and save time for immediate response. The system also focuses in providing payment methods for the booking, allowing the customers to pay through online using bank transfer and e-wallet which will be a convenient transaction process for them. Lastly, it also provides a content management system wherein staff and admin can add, remove, and modify travel and tour packages.

In summary, Travex aims to help and address the problems and needs of M&RC Travel and Tours by simplifying their operations of managing users, bookings, and transactions. By centralizing these processes into an online platform wherein the agency can effectively market its services while leveraging user ratings and reviews for strategic improvements and decisions. Furthermore, the system enhances customer experience by providing convenient and accessible portal wherein they can experience flexible booking options and receive helpful recommendations that streamlines the task of browsing and booking for the services of the travel agency.

2. OBJECTIVES

State the general objective and at least 5 specific objectives of the study

General Objectives:

To develop a Package Booking, Smart Recommendation and Customer Management System that will benefit a travel and tour agency by digitalizing their business operations and provides accessible services and packages for their clients.

Specific Objectives:

1. To eliminate the traditional process of manual booking through store front processes by implementing and digitalizing booking operations online.
2. To enhance the capability of the travel and tour agency in managing their bookings, users, and transactions through a management system.
3. To provide a flexible booking options and package customization for customers when it comes to browsing, checking, and booking travel and tour packages.
4. To enhance the service of the travel and tour agency and help them make informed decisions by allowing user ratings and reviews of travel and tour packages.
5. To enhance customer experience by implementing a smart recommendation system that shows personalized travel and tour packages to customers.

6. To evaluate the system functionality, usability, and reliability by conducting system quality testing and user acceptance testing in terms of ease of use, functionality and navigation.

3. SIGNIFICANCE OF THE STUDY
Discuss the importance of the project once developed

The Package Booking, Smart Recommendation and Customer Management System aims to streamline workflows, reduce manual workload and minimize human error by digitalizing business operations and processes of a travel and tour agency. The system will also provide a platform wherein customers can browse and book travel and tour packages conveniently.

For the Travel and Tour Agency
The system will be reducing manual booking errors and administrative workloads. By digitalizing business operations and booking processes, the agency will be able to manage their resources better and enhance their operations efficiently. This also gives the agency a competitive advantage by being able to enhance productivity and efficiency in marketing and providing their services.

For the Customers
The customers will benefit by having a platform for browsing and booking travel and tour packages benefiting from faster data processing, booking and payment transactions, improved customer service, personalized recommendations, and flexible booking options. This improves customer satisfaction by having a convenient approach in availing the services and products of the travel and tour agency, establishing easy access and eliminating traditional booking processes.

For Agency Administrators and Staffs
The administrators and staffs will be benefiting from the system by automating booking workflows and having an efficient panel wherein they can manage users, bookings, and transactions. Being able to add, remove, and modify travel and tour packages, which makes it convenient for administrators and staffs to monitor users, bookings and transaction details and status, simplifying day-to-day tasks and operations.

For Future Developers and Researchers
This study will serve as a reference for future similar system development projects. Showcasing practical application of system analysis and information system development in solving real-world problems. Innovating and improving current systems by integrating new processes and methods that will enhance business and organization operations. This project contributes to the field of information system development which can be a useful reference for future research and project development.

4. SCOPE AND DELIMITATION
Refers to the scope and inclusive frame/s of references as well as delimitation/s of the study

Scope of the System

The project team will be focusing in developing and implementing features that a Package Booking, Smart Recommendation and Customer Management System should have in order to streamline booking processes for customers and business operations within the travel and tour agency.

- **User Management System -** Users will be able to create accounts through registration, which will be used for logging in to the system. Account creation is implemented so that users can manage their profile and booking history, to eliminate the process of filling up forms when booking a travel and tour package.

- **Tour Package Display Catalogue Module with Flexible Booking Options -** The users or customers will be able to view the full details of the travel and tour package such as its price,

inclusion, exclusion, itineraries, available dates, and general information. The customers will experience flexible booking by being able to modify and avail custom itineraries in a specific package. The users can enhance their package by adding optional services such as add-on and upgrades in terms of accommodation, meals, tour options which showcases flexibility in choosing and booking for travel and tour packages.

- **Online Payment Transaction System -** Users have the capability to pay online through bank transfer and e-wallet transaction methods as they book themselves a travel and tour package.

- **Email Notifications -** Successful transactions, booking approvals, and other types of notifications will be sent through an email that is provided by the user upon registering an account.

- **Chatbot for Customer Support -** This chatbot will be mainly in charge of customer service, by entertaining inquiries related to the packages offered by the travel and tour agency. This chatbot can guide users by answering general FAQs and other specific questions about the agency.

- **Admin Panel and Dashboard for Generating Sales Reports and Analytics-** A main control panel for administrators and staff for viewing statistical data, wherein graphs and charts show booking trends, number of users, bookings and their status. Moreover, it will display insights and statistics regarding total booking, most booked packages, and revenues that will help the agency to make data-driven decisions, optimizing their travel and tour packages. This includes components such as managing user accounts, bookings, transaction details and status. Admins can see the list of user accounts and their details, tracking lists of completed booking and transactions. Lastly, the system will be able to generate reports that summarize sales performance and bookings which can be exported or printed on business documentation and auditing purposes.

- **Travel and Tour Package Content Management System -** This section is used by the staff and administrators for managing travel and tour packages, offering actions of adding, removing and modifying the details of the packages such as its inclusions, exclusions, itineraries, pricing and description.

- **Customer Invoice Module** – This allows the admin or staff to generate an invoice that will be sent to the customer when booking is confirmed. This invoice will display the total price of the package, the add-ons availed, the travel date, and the deadline for installation and full payments.

- **Customer Review and Rating System -** The capability of the customer to rate and review a travel and tour package will enhance the experiences of other customers by being able to view and read reviews from previous customers that have availed the tour package which helps them to make informed decisions.

- **Categorizing Travel and Tour Packages -** The travel and tour packages will be browsed and availed conveniently by customers in navigating travel and tour packages easily through categories such as types of activities, duration, budget, booking, countries, domestic or international travels.

- **Travel and Tour Package Wish list and Discounts Module -** This module handle calculating and applying discounts for the travel and tour packages which applies for PWD users, promo codes, and seasonal discounts. The users can add their packages to their wish list, wherein it will notify them whether the package is on sale or discount.

- **Smart Recommendation System -** This a feature that gives personalized travel and tour packages for the users by using hybrid filtering. This analyses the user preferences wherein upon making an account, the system will be asking for their preferences in categories, then it will proceed to showcase personalized recommendations to the customers.

- **VISA and Passport application service**– This feature allows users to upload files and requirements for VISA and Passport application which will be checked by the admin for verification and approval. It will guide users through a step-by-step process of availing VISA and passports, displaying status and sending notifications whether the step of the VISA and passport process is completed.

- **Rebooking and Cancellation System –** The system will be supporting this module that allows admin and users to rebook their existing reservations and submit cancellation request based on their policies. It also includes the time or date limitations on when customers can request cancellations and re-bookings.

Delimitation of the System

The project team will be delimiting features and other functions to ensure the technical and practical feasibility in developing the system.

- **Other Types of Booking -** The system will be focusing solely on the tour packages that the travel and tour agency offers, there will be no major modification when it comes to the number of days, and places indicated in the itinerary of the availed package. Further changes or customizations should be addressed physically in the travel agency office.

- **Limited Online Payment Methods -** The payment methods will only be limited to few available banks transfer and e-wallet companies.

- **Real time Checking of Flight and Hotels Availability -** The system does not include real time checking of the availability of the flight and hotels since travel and tour packages bundles includes transportation, accommodation, and tour activities.

- **Limitations for Smart Recommendation system** – The recommendation system will not be able to provide highly accurate recommendations for brand-new users due to lack of data and data sparsity such as wish listing, bookings and ratings of travel and tour packages.

- **Limitations for VISA and Passport application services** – In order to carefully verify the documents needed for the VISA and passport applications, the agency admin and staff will handle the verifications and reviewing of the submitted documents that will be processed for the VISA and passport services.

5. RELATED STUDIES

Present at most 5 studies to support your rationale and problems identified by the current systems. Note: Find the notable and published studies (either Scopus or ISI indexed papers)

The Travel and Tour Industry have evolved and transformed in the recent years as they are largely driven by advancements in information technology. Developing online travel and tour information systems have revolutionized how consumers or customers book and experience their travels. The implementation of these systems has shaped perceptions and intentions of consumers, becoming retaining customers due to the ease and accessibility of the services that the agency offers. With the integration of these information systems in tourism enterprises, it leads to agencies and businesses to thrive and prosper since it automates and digitalize its operations and processes, considering it as a competitive advantage from other agencies.

A Travel Booking and Management Application provides an end-to-end solution to the problems faced in planning trips, therefore it fulfils the requirements of travellers from transportation, accommodation and into

formulating and tailoring a complete travel and package for customers. Abu-AlShaeer et al., mentioned that this type of system aims to help customers in managing their bookings and at the same time benefitting the staff in keeping track of the online booking requests of the customers. It focuses on the purpose of eliminating the problems encountered in traditional processes of booking, developing a system that is suitable for web and mobile platform offering wide and convenient access for customers [1].

Gosela and Encarnacion indicated that travel and tour information systems should have a capability to offer flexible booking policies and options as this can attract broader range of tourist, moreover, implementing user reviews will help other tourists to make informed decisions based on feedback. Offering or displaying curated tour packages that aims to highlight local attractions and products will boost tourism and local businesses highlighted in the package. Moreover, offering flexible booking policies such as discounts can attract more customers while applying AI to provide personalized recommendations can enhance user experience [2].

While Gomes et al. suggested that travel and tour information systems should be able to target various groups of people, through getting or acquiring significant reviews from the experiences of others. Moreover, current systems should have admin control, including management of records, transactions, bookings, tour packages and other relevant data. Other travel and tour information systems focus more on availing and booking accommodation rather than a complete package with detailed itineraries [3].

In another study by Anik et al. and Jain and Vishali, they suggested that the scalability and extensibility of the travel and tour booking, and management system can be explored by accommodating emerging technologies such as integrating chatbots, which will be useful for enhancing customer service, by being able to address user inquiries and concerns immediately. Being able to interact with the users, in terms of answering FAQs and services provided by the travel and tour agency. Lastly, extending online payment options through popular banking and e-wallet methods enhances the experience of the users in booking and performing payment transactions [4] [5].

These studies showcased the development of travel and tour booking and management systems wherein it aims to enhance efficiency, reduce errors, and increase accessibility, supporting the need for a digitalized approach for travel and tour agencies. These studies also provided recommendations for improving the future system scope and functionalities of the system such as improving the booking flexibility for travel and tour packages, adding features that enhances customer experience such as review and ratings. These studies show areas that needs to be improved in other to enhance the efficiency and productivity of the system, benefiting customers and target agencies.

6. REFERENCES

Use IEEE citation and referencing style

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[5] L. Jain and R. Vishali, "Next-Gen MERN-Based Travel Booking Platform," International Research Journal of Innovations in Engineering and Technology (IRJIET), vol. 9, no. 4, pp. 188–191, 2025, doi: https://doi.org/10.47001/IRJIET/2025.904029.

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